

Questions

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Q1 - 2024 (04 Apr Shift 1)

Let a unit vector which makes an angle of 60° with $2\hat{i} + 2\hat{j} + \hat{k}$ and angle 45° with $\hat{i} - \hat{k}$ be $\vec{C}$. Then

$\vec{C} + \left(-\frac{1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$ is :

- (1) $\frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$
- (2) $\left(\frac{1}{\sqrt{3}} + \frac{1}{2}\right)\hat{i} + \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{2}}\right)\hat{j} + \left(\frac{1}{\sqrt{3}} + \frac{\sqrt{2}}{3}\right)\hat{k}$
- (3) $\frac{\sqrt{2}}{3}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{1}{2}\hat{k}$
- (4) $-\frac{\sqrt{2}}{3}\hat{i} + \frac{\sqrt{2}}{3}\hat{j} + \left(\frac{1}{2} + \frac{2\sqrt{2}}{3}\right)\hat{k}$

Q2 - 2024 (04 Apr Shift 1)

Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\vec{AB} = \hat{i} + 2\hat{j} - 7\hat{k}$, $\vec{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\vec{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}$, $d > 0$. Then the square of the length of the largest side of the triangle ABC is _____

Q3 - 2024 (04 Apr Shift 2)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$, $x \in \mathbb{R}$.

If $\vec{d}$ is the unit vector in the direction of $\vec{b} + \vec{c}$ such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to

- (1) 11
- (2) 3
- (3) 9
- (4) 6

Q4 - 2024 (04 Apr Shift 2)

For $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are mutually perpendicular, then the value of $(14 \cos \theta)^2$ is equal to

- (1) 50
- (2) 40
- (3) 25

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(4) 20

Q5 - 2024 (05 Apr Shift 1)

Let $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$. If

$\vec{a} \cdot \vec{c} = 130$, then $\vec{b} \cdot \vec{c}$ is equal to _____

Q6 - 2024 (05 Apr Shift 2)

Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in \left[0, \frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to:

(1) 110

(2) 124

(3) 121

(4) 105

Q7 - 2024 (05 Apr Shift 2)

Let $\vec{a} = 2\hat{i} + 5\hat{j} - \hat{k}$, $\vec{b} = 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{c}$ be three vectors such that $(\vec{c} + \hat{i}) \times (\vec{a} + \vec{b} + \hat{i}) = \vec{a} \times (\vec{c} + \hat{i})$. If $\vec{a} \cdot \vec{c} = -29$, then $\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k})$ is equal to:

(1) 15

(2) 12

(3) 10

(4) 5

Q8 - 2024 (06 Apr Shift 1)

If $A(3, 1, -1)$, $B\left(\frac{5}{3}, \frac{7}{3}, \frac{1}{3}\right)$, $C(2, 2, 1)$ and $D\left(\frac{10}{3}, \frac{2}{3}, \frac{-1}{3}\right)$ are the vertices of a quadrilateral $ABCD$, then its area is

(1) $\frac{2\sqrt{2}}{3}$ (2) $\frac{5\sqrt{2}}{3}$

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(3) $2\sqrt{2}$

(4) $\frac{4\sqrt{2}}{3}$

Q9 - 2024 (06 Apr Shift 1)

Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to _____

Q10 - 2024 (06 Apr Shift 2)

Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = ((\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i}) \times \hat{i}$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is :

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) 2

(4) $\frac{1}{5}$

Q11 - 2024 (06 Apr Shift 2)

Let $\vec{a} = 6\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $|\vec{c}| \geq 6$, $\vec{a} \cdot \vec{c} = 6|\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 60° , then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to:

(1) $\frac{9}{2}(6 - \sqrt{6})$

(2) $\frac{3}{2}\sqrt{6}$

(3) $\frac{9}{2}(6 + \sqrt{6})$

(4) $\frac{3}{2}\sqrt{3}$

Q12 - 2024 (08 Apr Shift 1)

The set of all α , for which the vectors $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$, is

(1) $\left(-\frac{4}{3}, 1\right)$

(2) $[0, 1)$

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(3) $\left(-\frac{4}{3}, 0\right]$

(4) $(-2, 0]$

Q13 - 2024 (08 Apr Shift 1)

Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____

Q14 - 2024 (08 Apr Shift 2)

Let $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 11\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = \vec{c} \times (-2\vec{a} + 3\vec{b})$. If $(2\vec{a} + 3\vec{b}) \cdot \vec{c} = 1670$, then $|\vec{c}|^2$ is equal to :

(1) 1609

(2) 1618

(3) 1600

(4) 1627

Q15 - 2024 (08 Apr Shift 2)

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r} \cdot \vec{a} = 3$, then 3λ is equal to:

(1) 21

(2) 30

(3) 25

(4) 27

Q16 - 2024 (09 Apr Shift 1)

Let three vectors $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ form a triangle such that $\vec{c} = \vec{a} - \vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to:

(1) 16

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(2) 14

(3) 12

(4) 10

Q17 - 2024 (09 Apr Shift 1)

Let $\vec{OA} = 2\vec{a}$, $\vec{OB} = 6\vec{a} + 5\vec{b}$ and $\vec{OC} = 3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\vec{OA}$ and $\vec{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral $OABC$ is equal to :

(1) 32

(2) 40

(3) 38

(4) 35

Q18 - 2024 (09 Apr Shift 2)

Between the following two statements:

Statement I : Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a} \times \vec{r} = \vec{a} \times \vec{b}$ and $\vec{a} \cdot \vec{r} = 0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC , $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$.

(1) Statement I is incorrect but Statement II is correct.

(2) Both Statement I and Statement II are correct.

(3) Statement I is correct but Statement II is incorrect.

(4) Both Statement I and Statement II are incorrect.

Q19 - 2024 (09 Apr Shift 2)

Let $\vec{a} = 2\hat{i} + \alpha\hat{j} + \hat{k}$, $\vec{b} = -\hat{i} + \hat{k}$, $\vec{c} = \beta\hat{j} - \hat{k}$, where α and β are integers and $\alpha\beta = -6$. Let the values of the ordered pair (α, β) , for which the area of the parallelogram of diagonals $\vec{a} + \vec{b}$ and $\vec{b} + \vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1, β_1) and (α_2, β_2) . Then $\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2$ is equal to

(1) 19

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Answer Key

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Solutions

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Q1

$$\vec{C} = C_1 \hat{i} + C_2 \hat{j} + C_3 \hat{k}$$

$$C_1^2 + C_2^2 + C_3^2 = 1$$

$$\vec{C} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = |C|\sqrt{9} \cos 60^\circ$$

$$2C_1 + 2C_2 - C_3 = \frac{3}{2}$$

$$C_1 - C_3 = 1$$

$$C_1 + 2C_2 = \frac{1}{2}$$

$$C_1 = \frac{\sqrt{2}}{3} + \frac{1}{2}$$

$$C_2 = \frac{-1}{3\sqrt{2}}$$

$$C_3 = \frac{\sqrt{2}}{3} - \frac{1}{2}$$

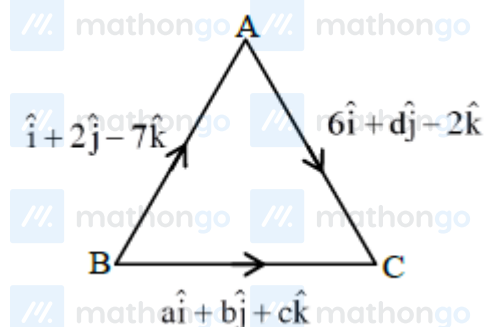
Q2

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Solutions

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$$\text{Area} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -7 \\ 6 & d & -2 \end{vmatrix} = 15\sqrt{2}$$

$$(-4 + 7d)\hat{i} - \hat{j}(-2 + 42) + \hat{k}(d - 12)$$

$$(7d - 4)^2 + (40)^2 + (d - 12)^2 = 1800$$

$$50d^2 - 80d - 40 = 0$$

$$5d^2 - 8d - 4 = 0$$

$$5d^2 - 10d - 2d - 4$$

$$5d(d - 2) + 2(d - 2) = 0$$

$$d = 2 \text{ or } d = -\frac{2}{5}$$

$$\therefore d > 0, d = 2$$

$$(a + 1)\hat{i} + (b + 2)\hat{j} + (c - 7)\hat{k} = 6\hat{i} + 2\hat{j} - 2\hat{k}$$

$$a + 1 = 6 \text{ and } b + 2 = 2, c - 7 = -2$$

$$a = 5 \quad b = 0 \quad c = 5$$

$$|AB| = \sqrt{1 + 4 + 49} = \sqrt{54}$$

$$|BC| = \sqrt{25 + 25} = \sqrt{50}$$

$$|AC| = \sqrt{86 + 4 + 4} = \sqrt{44}$$

Ans. 54

Q3

$$\vec{d} = \lambda(\vec{b} + \vec{c})$$

$$\vec{a} \cdot \vec{d} = \lambda(\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$$

$$1 = \lambda(1 + x + 5)$$

$$1 = \lambda(x + 6) \dots (1)$$

Solutions

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$$|\vec{d}| = 1$$

$$|\lambda(\vec{b} + \vec{c})| = 1$$

$$|\lambda((x+2)\hat{i} + 6\hat{j} - 2\hat{k})| = 1$$

$$\lambda^2((x+2)^2 + 6^2 + 2^2) = 1$$

$$x^2 + 4x + 4 + 36 + 4 = (x+6)^2$$

$$x^2 + 4x + 44 = x^2 + 12x + 36$$

$$8x = 8, x = 1$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & -5 \\ x & 2 & 3 \end{vmatrix} = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\begin{vmatrix} 0 & 0 & 1 \\ -2 & 9 & -4 \\ x-2 & -1 & 3 \end{vmatrix} = 2 - 9(x-2)$$

$$= 20 - 9x$$

$$\text{at } x = 1$$

$$20 - 9 = 11$$

Q4

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0, \lambda > 0$$

$$|\vec{a}|^2 - |\vec{b}|^2 = 0 \rightarrow 1 + \lambda^2 + 9 = 9 + 1 + 4$$

$$\therefore \lambda = 2, \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \cdot |\vec{b}|} = \frac{3 - \lambda - 6}{\sqrt{14} \cdot \sqrt{14}}$$

$$14 \cos \theta = 3 - 8 = -5$$

$$\therefore (14 \cos \theta)^2 = 25$$

Q5

$$(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$$

$$(2\vec{b} + 4\vec{a}) \times \vec{c} = 0$$

$$\vec{c} = \lambda(4\vec{a} + 2\vec{b}) = \lambda(8\hat{i} - 14\hat{j} + 30\hat{k})$$

$$\vec{a} \cdot \vec{c} = 130$$

$$8\lambda + 42\lambda + 210\lambda = 130$$

$$\lambda = \frac{1}{2}$$

$$\vec{c} = 4\hat{i} - 7\hat{j} + 15\hat{k}$$

$$\vec{b} \cdot \vec{c} = 8 + 7 + 15 = 30$$

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Solutions

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Q6

$$|\vec{c} - \vec{a}| = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a} \cdot \vec{c}$$

$$= |\vec{c}|^2 + 4 - 0$$

$$\therefore \vec{a} = \vec{b} \times \vec{c}$$

$$|\vec{a}| = |\vec{b} \times \vec{c}|$$

$$2 = 3|\vec{c}| \sin \alpha$$

$$|\vec{c}| = \frac{2}{3} \operatorname{cosec} \alpha \quad \alpha \in \left[0, \frac{\pi}{3}\right]$$

$$|\vec{c}|_{\min} = \frac{2}{3} \times \frac{2}{\sqrt{3}} \operatorname{cosec} \alpha \in \left[\frac{2}{\sqrt{3}}, \infty\right)$$

$$\Rightarrow 27|\vec{c} - \vec{a}|_{\min}^2 = 27\left(\frac{16}{27} + 4\right) = 124$$

Q7

Let's assume

$$\vec{v} = \vec{a} + \vec{b} + \hat{i}$$

$$= 5\hat{i} + 3\hat{j} + \hat{k}$$

$$\text{and } \vec{c} + \hat{i} = \vec{p}$$

So,

$$\vec{p} \times \vec{v} = \vec{a} \times \vec{p}$$

$$\vec{p} \times \vec{v} + \vec{p} \times \vec{a} = \vec{0}$$

$$\vec{p} \times (\vec{v} + \vec{a}) = \vec{0}$$

$$\Rightarrow \vec{p} = \lambda(\vec{v} + \vec{a})$$

$$\vec{c} + \hat{i} = \lambda(7\hat{i} + 8\hat{j})$$

$$\vec{a} \cdot \vec{c} + \vec{a} \cdot \hat{i} = \lambda \vec{a} \cdot (7\hat{i} + 8\hat{j})$$

$$-29 + 2 = \lambda(14 + 40)$$

$$\lambda = -\frac{1}{2}$$

$$\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k}) + \hat{i} \cdot (-2\hat{i} + \hat{j} + \hat{k}) = \lambda(7\hat{i} + 8\hat{j}) \cdot (-2\hat{i} + \hat{j} + \hat{k})$$

$$= -\frac{1}{2}(-14 + 8) + 2 = 5$$

Q8

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Solutions

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$$\text{Area} = \frac{1}{2} |\overrightarrow{BD} \times \overrightarrow{AC}|$$

$$\overrightarrow{BD} = \frac{5}{3}\hat{i} - \frac{5}{3}\hat{j} - \frac{2}{3}\hat{k}$$

$$\overrightarrow{AC} = \hat{i} - \hat{j} - 2\hat{k}$$

Q9

$$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{c} = (1, 8, 13)$$

$$\vec{a} \times (\vec{a} \times \vec{b}) + \vec{a} \times (\vec{a} \times \vec{c}) + \vec{a} \times (\vec{b} \times \vec{c})$$

$$= \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$(\vec{a} \cdot \vec{b})\vec{a} - a^2\vec{b} + (\vec{a} \cdot \vec{c})\vec{a} - a^2\vec{c} + (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$\Rightarrow -26\vec{a} - 29\vec{b} + 13\vec{a} - 29\vec{c} + 13\vec{b} + 26\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$\Rightarrow -13\vec{a} - 16\vec{b} - 3\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$\Rightarrow -13\vec{a} \cdot \vec{b} - 16b^2 - 3\vec{b} \cdot \vec{c} = \{\vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})\} \cdot \vec{b}$$

$$\Rightarrow (-13)(-26) - 16(50) - 3\vec{b} \cdot \vec{c} = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 8 & 13 \\ 3 & 4 & -5 \end{vmatrix}$$

$$\Rightarrow -462 - 3\vec{b} \cdot \vec{c} = -396$$

$$\Rightarrow \vec{b} \cdot \vec{c} = -22$$

$$\text{Hence } 24 - \vec{b} \cdot \vec{c} = 46$$

Q10

$$\vec{a} \times (\hat{i} + \hat{j}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= \hat{i} - \hat{j} + \hat{k}$$

$$(\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i} = \hat{k} + \hat{j}$$

$$((\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i}) \times \hat{i} = \hat{j} - \hat{k}$$

$$\text{projection of } \vec{a} \text{ on } \hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$= \frac{1+1}{\sqrt{2}} = \sqrt{2}$$

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Solutions

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Q11

$$|(\vec{a} \times \vec{b} \times \vec{c})| = |\vec{a} \times \vec{b}| |\vec{c}| \frac{\sqrt{3}}{2}$$

$$|\vec{c} - \vec{a}| = 2\sqrt{2}$$

$$|\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a} = 8$$

$$|z|^2 + 38 - 12|z| = 8$$

$$|z|^2 - 12|z| + 30 = 0$$

$$|z| = \frac{12 \pm \sqrt{144 - 120}}{2}$$

$$= \frac{12 \pm 2\sqrt{6}}{2}$$

$$|z| = 6 + \sqrt{6}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$\hat{i} - \hat{j} + 5\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{27}$$

$$|(\vec{a} \times \vec{b}) \times \vec{z}| = \sqrt{27}(6 + \sqrt{6}) \frac{\sqrt{3}}{2}$$

$$\frac{9}{2}(6 + \sqrt{6})$$

Q12

$$\vec{a} = \alpha \hat{i} + 6\hat{j} - 3\hat{k}$$

$$\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha\hat{k}$$

$$\text{so } \vec{a} \cdot \vec{b} < 0, \forall t \in \mathbb{R}$$

$$\alpha t^2 - 12 + 6\alpha t < 0$$

$$\alpha t^2 + 6\alpha t - 12 < 0, \forall t \in \mathbb{R}$$

$$\alpha < 0, \text{ and } D < 0$$

$$36\alpha^2 + 48\alpha < 0$$

$$12\alpha(3\alpha + 4) < 0$$

$$-\frac{4}{3} < \alpha < 0$$

$$\text{also for } \alpha = 0, \vec{a} \cdot \vec{b} < 0$$

$$\text{hence } \alpha \in \left(-\frac{4}{3}, 0\right]$$

Q13

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Solutions

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$$\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$$

$$\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$$

$$\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{b} + \vec{c} = 20\hat{i} + 5\hat{j} - 12\hat{k}$$

$$\vec{b} - \vec{c} = -14\hat{i} + 9\hat{j} - 14\hat{k}$$

$$(\vec{r} - (\vec{b} + \vec{c})) \times \vec{a} = 0$$

$$\vec{r} - (\vec{b} + \vec{c}) = \lambda \vec{a}$$

$$\vec{r} = \lambda \vec{a} + \vec{b} + \vec{c}$$

$$\text{But } \vec{r} \cdot (\vec{b} - \vec{c}) = 0$$

$$\Rightarrow (\lambda \vec{a} + \vec{b} + \vec{c}) \cdot (\vec{b} - \vec{c}) = 0$$

$$\Rightarrow \lambda \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{b} - \lambda \vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{c} = 0$$

$$\lambda = \frac{\vec{c} \cdot \vec{c} - \vec{b} \cdot \vec{b}}{\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c}} = \frac{294 - 227}{-389} = \frac{-67}{593}$$

$$\therefore \vec{r} = \vec{b} + \vec{c} - \frac{67}{593} \vec{a}$$

$$\Rightarrow 593 \vec{r} + 67 \vec{a} = 593(\vec{b} + \vec{c})$$

$$\Rightarrow |\vec{b} + \vec{c}|^2 = 569$$

Q14

$$(\vec{a} + \vec{b}) \times \vec{c} - \vec{c} \times (-2\vec{a} + 3\vec{b}) = 0$$

$$(\vec{a} + \vec{b}) \times \vec{c} + (-2\vec{a} + 3\vec{b}) \times \vec{c} = 0$$

$$\Rightarrow (\vec{a} + \vec{b}) - 2\vec{a} + 3\vec{b} \times \vec{c} = 0$$

$$\Rightarrow \vec{c} = \lambda(4\vec{b} - \vec{a})$$

$$\Rightarrow \lambda(44\hat{i} - 4\hat{j} + 4\hat{k} - 4\hat{i} + \hat{j} - \hat{k})$$

$$= \lambda(40\hat{i} - 3\hat{j} + 3\hat{k})$$

Now

$$(8\hat{i} - 2\hat{j} + 2\hat{k} + 33\hat{i} - 3\hat{j} + 3\hat{k}) \cdot \lambda(40\hat{i} - 3\hat{j} + 3\hat{k}) = 1670$$

$$\Rightarrow (41\hat{i} - 5\hat{j} + 5\hat{k}) \cdot (40\hat{i} - 3\hat{j} + 3\hat{k}) \times \lambda = 1670$$

$$\Rightarrow (1640 + 15 + 15)\lambda = 1670 \Rightarrow \lambda = 1$$

$$\text{so } \vec{c} = 40\hat{i} - 3\hat{j} - 3\hat{k}$$

$$\Rightarrow |\vec{c}|^2 = 1600 + 9 + 9 = 1618$$

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Solutions

MathonGo

Q15

$$\vec{r} = k(\vec{b} + \vec{c})$$

$$\vec{r} \cdot \vec{a} = 3$$

$$\vec{r} \cdot \vec{a} = k(\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$$

$$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$$

$$3 = k(-6 + 3\lambda) \dots (1)$$

$$\vec{r} = k(5\hat{i} + 2\hat{j} - (5 - \lambda)\hat{k})$$

$$|\vec{r}| = k\sqrt{25 + 4 + 25 + \lambda^2 - 10\lambda} = 1 \dots (2)$$

$$k = \frac{3}{-6+3\lambda} = \frac{1}{-2+\lambda} \quad \text{put in (2)}$$

$$4 + \lambda^2 - 4\lambda = 54 + \lambda^2 - 10\lambda$$

$$6\lambda = 50$$

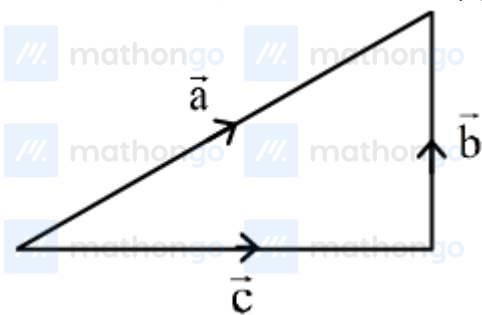
$$3\lambda = 25$$

Q16

$$\vec{c} = \vec{a} - \vec{b}$$

$$\Rightarrow (x, y, z) = (\alpha - 5, 1, -2)$$

$$\Rightarrow x = \alpha - 5, y = 1, z = -2 \dots (1)$$



$$\text{Area of } \Delta = 5\sqrt{6} \text{ (given)}$$

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Solutions

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$$\frac{1}{2} |\vec{a} \times \vec{c}| = 5\sqrt{6}$$

$$\left\| \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 4 & 2 \\ x & 1 & -2 \end{matrix} \right\| = 10\sqrt{6}$$

$$\Rightarrow |-10\hat{i} - \hat{j}(-2\alpha - 2x) + \hat{k}(\alpha - 4x)| = 10\sqrt{6}$$

$$\Rightarrow (2\alpha + 2\alpha - 10)^2 + (\alpha - 4\alpha + 20)^2 = 500$$

$$\Rightarrow (4\alpha - 10)^2 + (20 - 3\alpha)^2 = 500$$

$$\Rightarrow 25\alpha^2 - 80\alpha - 120\alpha = 0$$

$$\Rightarrow \alpha(25\alpha - 200) = 0$$

$$\Rightarrow \alpha = 8 \text{ (given } \alpha \text{ is +ve number)}$$

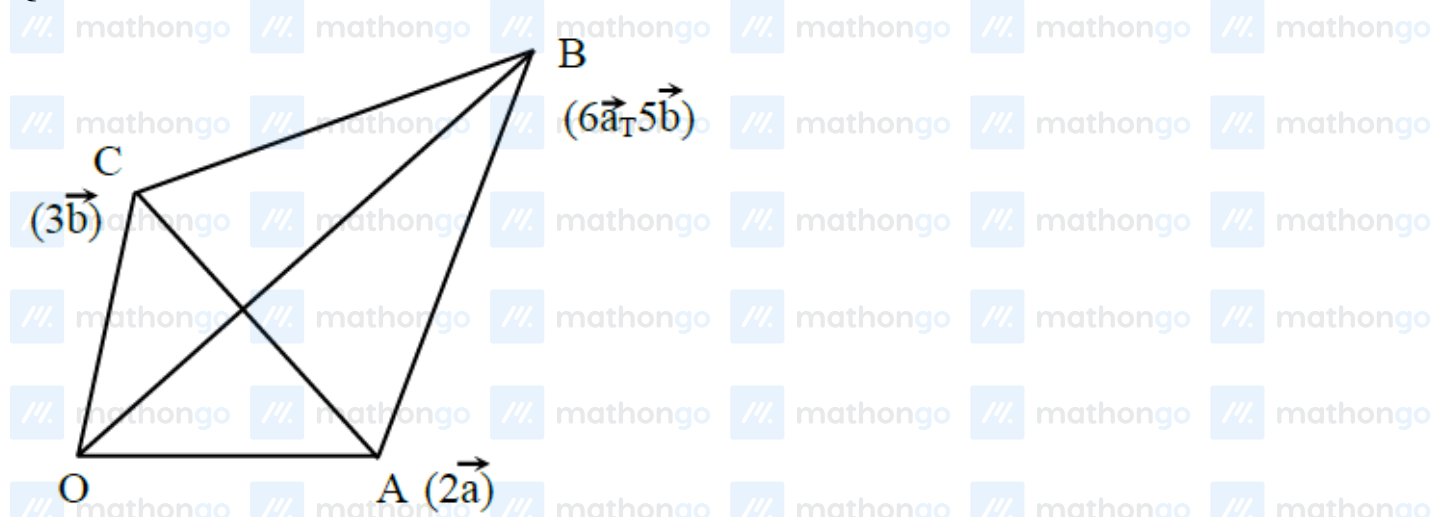
$$\Rightarrow x = \alpha - 5 = 3$$

$$|\vec{c}|^2 = x^2 + y^2 + z^2$$

$$= 9 + 1 + 4$$

$$= 14$$

Q17



Area of parallelogram having sides

$$\vec{OA} \& \vec{OC} = |\vec{OA} \times \vec{OC}| = |2\vec{a} \times 3\vec{b}| = 15$$

$$6|\vec{a} \times \vec{b}| = 15$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \frac{5}{2} \dots \dots \dots (1)$$

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Solutions

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Area of quadrilateral

$$\begin{aligned}
 \text{OABC} &= \frac{1}{2} \left| \vec{d}_1 \times \vec{d}_2 \right| \\
 &= \frac{1}{2} |\vec{AC} \times \vec{OB}| = \frac{1}{2} |(3\vec{b} - 2\vec{a}) \times (6\vec{a} + 5\vec{b})| \\
 &= \frac{1}{2} |18\vec{b} \times \vec{a} - 10\vec{a} \times \vec{b}| = 14 |\vec{a} \times \vec{b}| \\
 &= 14 \times \frac{5}{2} = 35
 \end{aligned}$$

Q18

$$\begin{aligned}
 \vec{a} &= \hat{i} + 2\hat{j} - 3\hat{k} \\
 \vec{a} &= 2\hat{i} + \hat{j} - \hat{k} \\
 \vec{a} \times \vec{r} &= \vec{a} \times \vec{b}; \vec{a} \cdot \vec{r} = 0 \\
 \Rightarrow \vec{a} \times (\vec{r} - \vec{b}) &= \vec{0} \\
 \Rightarrow \vec{a} &= \lambda(\vec{r} - \vec{b})
 \end{aligned}$$

$$\begin{aligned}
 \vec{a} \cdot \vec{a} &= \lambda(\vec{a} \cdot \vec{r} - \vec{a} \cdot \vec{b}) \\
 14 &= -7\lambda \Rightarrow \lambda = -2
 \end{aligned}$$

$$\begin{aligned}
 \frac{-\vec{a}}{2} &= \vec{r} - \vec{b} \Rightarrow \vec{r} = \vec{b} - \frac{\vec{a}}{2} \\
 &= \frac{2\vec{b} - \vec{a}}{2} = \frac{3\hat{i} + \hat{k}}{2}
 \end{aligned}$$

Statement (I) is incorrect

$$\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$$

$$2A + 2B + 2C = 2\pi$$

$$\begin{aligned}
 \cos 2A + \cos 2B + \cos 2C \\
 = -1 - 4 \cos A \cdot \cos B \cdot \cos C
 \end{aligned}$$

$$\geq -1 - 4 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$= -\frac{3}{2}$$

Statement (II) is correct.

Q19

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Solutions

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$$\text{Area of parallelogram} = \frac{1}{2} \left| \vec{d}_1 \times \vec{d}_2 \right|$$

$$A = \frac{1}{2} |(\vec{a} + \vec{b}) \times (\vec{b} + \vec{c})| = \frac{\sqrt{21}}{2}$$

$$\text{so, } \vec{a} + \vec{b} = \hat{i} + \alpha\hat{j} + 2\hat{k}$$

$$\vec{b} + \vec{c} = -\hat{i} + \beta\hat{j}$$

$$(\vec{a} + \vec{b}) \times (\vec{b} + \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha & 2 \\ -1 & \beta & 0 \end{vmatrix}$$

$$= \hat{i}(-2\beta) - \hat{j}(2) + \hat{k}(\beta + \alpha)$$

$$|(\vec{a} + \vec{b}) \times (\vec{b} + \vec{c})| = \sqrt{4\beta^2 + 4 + (\alpha + \beta)^2} = \sqrt{21}$$

$$4\beta^2 + 4 + \alpha^2 + \beta^2 + 2\alpha\beta = 21$$

$$\alpha^2 + 5\beta^2 - 12 = 17$$

$$\alpha^2 + 5\beta^2 = 29$$

$$\text{and } \alpha\beta = -6$$

and given α, β are integers

$$\text{so, } \alpha = -3, \beta = 2$$

or

$$\alpha = 3, \beta = -2$$

$$(\alpha_1, \beta_1) = (-3, 2)$$

$$(\alpha_2, \beta_2) = (3, -2)$$

$$\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2 = 9 + 4 + 6 = 19$$

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